DUAL-MOTOR PMSM DYNAMOMETER PLATFORM
FUNCTIONAL REQUIREMENTS & VALIDATION

TI C2000 F28069M dual-inverter control, interfaces, verification, and acceptance

[redacted]	Herder Elektronische Systemen
Document scope	System behavior, operating modes, external interfaces, safeguards, verification, and acceptance
Evidence basis	Editable Simulink models, custom ADC source, CAN interfaces, scripts, schematics, and retained verification evidence
Distribution	Customer-facing; selected source models and engineering evidence included

DOCUMENT PURPOSE
Defines the controlled functional, performance, interface, safety, verification, and acceptance requirements for the dual-motor PMSM dynamometer platform. Detailed implementation and efficiency characterization are controlled in the companion engineering report.

WHY THIS IS AN FRD
This document defines what the released platform shall do, the interfaces it shall preserve, the evidence required to verify compliance, and the acceptance conditions for release. It is not an implementation walkthrough or a results report.

Functional Requirements Document Framework

Purpose, Scope, and System Boundary

This FRD governs the released dual-motor dynamometer platform at the level of externally observable behavior, interfaces, safeguards, evidence obligations, and acceptance. The companion Embedded Control & Characterization Report explains how the Simulink, CAN, ADC, logging, and analysis implementations satisfy this controlled requirement set.

Users and Operational Scenarios

User / scenario	Decision supported
Controls engineer	Modify target or host models without violating controlled interfaces or operating behavior.
Test engineer	Select verification methods, evidence, and acceptance criteria for a release.
Customer or reviewer	Understand system capability, boundaries, and demonstrated compliance.
Maintainer	Determine whether a proposed change requires FRD revision and re-verification.

Requirement Conventions

Mandatory requirements use “shall.” Each requirement has a unique identifier, verification method, acceptance basis, and release status. Descriptive text explains intent but does not replace the requirement statement.

Status	Meaning
Verified	Objective evidence has been inspected and accepted.
Demonstrated	Behavior was observed on the retained bench evidence set.
Open	Evidence or qualification remains incomplete.
Not claimed	Capability is intentionally outside the public release boundary.

1. System Role and Delivered Capability

The platform couples two encoder-equipped permanent-magnet synchronous machines on a common shaft. Motor 1 acts as the drive machine and establishes shaft speed. Motor 2 applies opposing or assisting q-axis current to create a controllable load. A TI C2000 F28069M target executes the real-time motor-control functions; host models provide supervisory commands, acquisition, and test orchestration.

- Embedded target retains ownership of high-rate current and speed control.
- Supervisory host supplies speed, load, enable, selection, and diagnostic commands.
- Independent encoder channels preserve separate feedback for both machines.
- CAN and host acquisition paths support repeatable command and evidence workflows.
- Custom ADC source extends analog acquisition while generated build output remains excluded.

1.1 Normal Operating Sequence

A normal run begins with wiring and polarity inspection, low-risk deployment, phase-order and encoder-direction confirmation, low-speed enable, controlled speed/load application, data capture, and post-run evidence review. The host coordinates the sequence but does not become the real-time control clock.

1.2 Operating Modes

Mode	Required behavior
Disabled / safe	Commanded torque production is inhibited and the system remains ready for inspection or configuration.
Drive-only	Motor 1 regulates shaft speed while Motor 2 remains at zero or bounded load command.
Loaded dynamometer	Motor 1 regulates speed while Motor 2 applies bounded opposing or assisting q-axis current.
CAN-supervised	Command words are received over CAN and mapped into existing control references and diagnostics.
Acquisition / review	Signals are captured and retained for verification and characterization without altering embedded timing.

2. Functional Requirements

The following requirements define required behavior independent of the internal block arrangement. A compliant implementation may evolve, but these externally visible functions shall remain intact unless this FRD is revised.

ID	Requirement	Verification	Status
SYS-001	The embedded target shall execute the fast motor-control loops independently of host display or logging timing.	Inspection + functional test	Demonstrated
SYS-002	The platform shall command one machine as the drive unit and the second as a controllable load or assist unit.	Functional test	Demonstrated
SYS-003	The platform shall accept a supervisory speed reference for the drive machine.	Functional test	Demonstrated
SYS-004	The platform shall accept a q-axis current or equivalent load reference for the load machine.	Functional test	Demonstrated
SYS-005	The platform shall provide explicit enable and disable control for commanded operation.	Inspection + test	Demonstrated
SYS-006	The platform shall maintain independent encoder	Inspection + feedback test	Demonstrated

	feedback paths for both machines.		
SYS-007	The released workflow shall support repeatable parameter setup, capture, and post-run review.	Procedure review	Verified

2.1 Requirement Rationale

SYS-001 prevents nondeterministic host refresh or file I/O from becoming part of the current-control timing chain. SYS-002 through SYS-006 preserve the dual-machine dynamometer role and the feedback separation required to diagnose either machine. SYS-007 establishes repeatability as a release obligation rather than an informal operator practice.

3. External Interface Requirements

Interface requirements control the boundaries most likely to cause hardware damage, command ambiguity, or invalid evidence: DC input, motor phases, encoders, CAN, deployment connection, and analog acquisition.

ID	Requirement	Verification	Status
IF-001	The inverter stack shall be supplied from 24 V DC with verified polarity and current limiting before enable.	DMM check + startup test	Demonstrated
IF-002	Each motor shall remain connected consistently to its assigned three-phase inverter output.	Continuity / wiring inspection	Demonstrated
IF-003	Motor 1 and Motor 2 encoder channels shall remain electrically and logically distinct.	Pin inspection + feedback test	Demonstrated
IF-004	The CAN bus shall provide CANH, CANL, reference ground, and appropriate termination.	Resistance + communication test	Demonstrated
IF-005	The target shall retain the debug/deployment interface required for configuration and diagnostics.	Connection inspection	Demonstrated
IF-006	The custom ADC integration shall configure the released simultaneous-sampling channels, trigger sources, acquisition windows, and end-of-conversion interrupt routing.	Source inspection	Verified

3.1 Controlled CAN Contract

Item	Public requirement
Command identifier	Standard CAN identifier 0x100.
Command payload	Eight-byte payload containing four 16-bit supervisory command words.
Return identifier	Standard CAN identifier 0x101 for command echo or diagnostic return.
Control integration	Decoded command words feed the existing reference, enable, selection, and FOC signal paths.

3.2 Controlled ADC Contract

SOC pair	Inputs	Trigger	Completion
SOC12 / SOC13	ADCINA2 / ADCINB2	ePWM1	EOC13 -> ADCINT5
SOC14 / SOC15	ADCINA1 / ADCINB1	ePWM1	EOC15 -> ADCINT9

The custom source establishes the additional analog acquisition path. Detailed register-level implementation and source excerpts are maintained in the companion engineering report and embedded-control folder.

4. Evidence and Data Requirements

The public release must preserve enough evidence to confirm required behavior without distributing private raw datasets, generated firmware, or vendor manufacturing packages.

ID	Requirement	Verification	Status
DAT-001	The release shall expose the speed, current, voltage, torque-related, power-related, and diagnostic quantities required by the approved verification workflow.	Model/script inspection	Demonstrated
DAT-002	Published evidence shall remain traceable to retained source models, scripts, or approved outputs.	Repository inspection	Verified
DAT-003	Measured evidence shall distinguish observed operating regions from unmeasured or unsupported regions.	Figure/report inspection	Verified
DAT-004	Interrupted or incomplete acquisition shall not be represented as complete evidence.	Procedure inspection	Verified
DAT-005	The public release shall exclude generated code,	Release inspection	Verified

	binaries, private raw data, and institution-specific instructional material.		

Efficiency methodology and maps are not part of this FRD. They are contained only in the Embedded Control & Characterization Report. This document controls the evidence obligations and acceptance basis, not the interpretation of characterization results.

5. Safety, Constraints, and Release Limitations

ID	Requirement	Verification	Status
SAFE-001	DC polarity shall be verified before inverter power is applied.	Inspection	Required
SAFE-002	Initial operation after wiring or software change shall begin at low speed and bounded load.	Procedure review	Required
SAFE-003	Software disable shall not be represented as an independent safety function.	Document review	Verified
SAFE-004	Rotating components shall be secured and kept clear during operation.	Procedure review	Required
REL-001	The release shall not claim calibrated metrology, production safety certification, EMC qualification, or long-duration thermal qualification.	Document review	Verified

5.1 Assumptions and Dependencies

- Compatible MATLAB/Simulink and C2000 support packages are available.
- Vendor controller and inverter documentation is used to verify board-revision pin assignments.
- Bench wiring, motor mounting, and coupler integrity are checked before operation.
- Instrumentation calibration remains outside the public release unless separately documented.

6. Verification and Acceptance Plan

Verification confirms that each requirement is satisfied by inspection, analysis, demonstration, or test. Validation confirms that the integrated platform supports its intended drive/load, supervisory-control, and evidence-generation use cases.

Method	Application
Inspection	Review source models, custom C source, wiring references, repository contents, and document statements.
Analysis	Evaluate equations, signal definitions, and consistency of the

	evidence workflow.
Demonstration	Observe integrated command, feedback, CAN, and acquisition behavior.
Test	Apply controlled speed/load commands and confirm required responses and interface operation.

6.1 Requirements Traceability Matrix

Requirement	Retained evidence	Status
SYS-001	Target model and deployment evidence	Demonstrated
SYS-002	Dual-machine target/host models	Demonstrated
SYS-003	Host reference path	Demonstrated
SYS-004	Load-command path	Demonstrated
SYS-005	Enable path	Demonstrated
SYS-006	Encoder interface evidence	Demonstrated
SYS-007	Setup and capture scripts	Verified
IF-001	Power-up procedure	Demonstrated
IF-002	Interface summary	Demonstrated
IF-003	Connector map and feedback test	Demonstrated
IF-004	CAN architecture and bus test	Demonstrated
IF-005	Deployment workflow	Demonstrated
IF-006	F28069M_ADC_Custom_Config.c	Verified
DAT-001	Models and logging scripts	Demonstrated
DAT-002	Repository evidence map	Verified
DAT-003	Companion characterization report	Verified
DAT-004	Acquisition procedure	Verified
DAT-005	NOTICE.md and ZIP inspection	Verified

6.2 Acceptance Basis

A release is acceptable when all mandatory requirements are verified or explicitly bounded as open, the repository passes release inspection, report references resolve to retained evidence, and no excluded private or institution-specific materials are present.

7. Document Control and Release Gate

Release check	Acceptance criterion
Requirement quality	Identifiers are unique; mandatory statements are atomic and use “shall.”
Traceability	Every mandatory requirement maps to a verification method and retained evidence.
Document separation	Efficiency characterization appears only in the companion engineering report.

Public boundary	Generated code, binaries, private raw data, and instructional material are excluded.
Visual QA	The DOCX and PDF are rendered and reviewed page by page.
Archive QA	The ZIP opens at repository root and passes integrity testing.

Revision 2.2 applies the actual controlled FRD template structure, cover, typography, headers, table system, and document-control hierarchy from the approved Dual-Core Optical Motion Testbed FRD package.